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INSULATION RESISTANCE DETECTION DEVICE AND INSULATION RESISTANCE DETECTION METHOD

Abstract

An insulation resistance detection device includes: a voltage measurer that measures a voltage at a first node among a plurality of nodes between a plurality of voltage dividing resistors included in an insulation detection circuit for detecting insulation resistance in a path through which a current from a battery flows; a determiner that determines whether the voltage at the first node measured by the voltage measurer is lower than or equal to a predetermined voltage; and a switcher that switches a state of a switch connected to a second node among the plurality of nodes to cause the voltage measured by the voltage measurer to exceed the predetermined voltage, when the voltage at the first node measured by the voltage measurer is determined to be lower than or equal to the predetermined voltage.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an insulation resistance detection device and an insulation resistance detection method for detecting insulation resistance in a path through which current from a battery flows.

BACKGROUND ART

[0002] Patent Literature (PTL) 1 describes a device that can detect insulation resistance in a path through which current from a battery flows.

CITATION LIST

Patent Literature

[0003] [PTL 1] Japanese Unexamined Patent Application Publication No. 2018-128454

SUMMARY OF INVENTION

Technical Problem

[0004] However, the device described in PTL 1 detects the insulation resistance by measuring the voltage at one node, and if the voltage measured at that node becomes low due to a drop in battery voltage or during the measurement control process, the measurement error of the voltage at that node will increase, and the detection accuracy of the insulation resistance will decrease.

[0005] Therefore, the present disclosure provides an insulation resistance detection device and the like that can suppress a decrease in the detection accuracy of insulation resistance even if the voltage of the node at which the voltage is measured decreases.

Solution to Problem

[0006] An insulation resistance detection device according to one embodiment of the present disclosure includes: a voltage measurer that measures a voltage at a first node among a plurality of nodes between a plurality of voltage dividing resistors connected between a positive terminal and a negative terminal of a battery, the plurality of voltage dividing resistors being included in an insulation detection circuit for detecting insulation resistance in a path through which a current from the battery flows; a determiner that determines whether the voltage at the first node measured by the voltage measurer is lower than or equal to a predetermined voltage; and a switcher that switches a state of a switch connected to a second node among the plurality of nodes to cause the voltage measured by the voltage measurer to exceed the predetermined voltage, when the voltage at the first node measured by the voltage measurer is determined to be lower than or equal to the predetermined voltage.

[0007] An insulation resistance detection method according to one embodiment of the present disclosure includes: measuring a voltage at a first node among a plurality of nodes between a plurality of voltage dividing resistors connected between a positive terminal and a negative terminal of a battery, the plurality of voltage dividing resistors being included in an insulation detection circuit for detecting insulation resistance in a path through which a current from the battery flows; determining whether the voltage at the first node measured in the measuring is lower than or equal to a predetermined voltage; and switching a state of a switch connected to a second node among the plurality of nodes to cause the voltage measured in the measuring to exceed the predetermined voltage, when the voltage at the first node measured in the measuring is determined to be lower than or equal to the predetermined voltage.

Advantageous Effects of Invention

[0008] According to an insulation resistance detection device according to one embodiment of the present disclosure, a decrease in the detection accuracy of the insulation resistance can be suppressed even if the voltage of the node at which the voltage is measured decreases.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a configuration diagram showing an example of an insulation resistance detection device according to Embodiment 1.

[0010] FIG. 2 is a diagram for explaining the effect of the insulation resistance detection device according to Embodiment 1.

[0011] FIG. 3 is a configuration diagram showing an example of an insulation resistance detection device according to a variation of Embodiment 1.

[0012] FIG. 4 is a configuration diagram showing an example of an insulation resistance detection device according to Embodiment 2.

[0013] FIG. 5 is a flowchart showing an example of an insulation resistance detection method for another embodiment.

DESCRIPTION OF EMBODIMENTS

[0014] Hereinafter, the embodiments will be described in detail with reference to the drawings.

[0015] It should be noted that each of the embodiments described below shows comprehensive or specific examples. The numerical values, shapes, materials, components, arrangement positions and connection forms of the components, and the like shown in the following embodiments are merely examples, and are not intended to limit the present disclosure.

Embodiment 1

[0016] Hereinafter, insulation resistance detection device 1 in Embodiment 1 will be described with reference to FIG. 1.

[0017] FIG. 1 is a configuration diagram showing an example of insulation resistance detection device 1 according to Embodiment 1. It should be noted that in addition to insulation resistance detection device 1, FIG. 1 also shows battery Bat, insulation resistance Riso1 and Riso2 in a path through which current from battery Bat flows, and insulation detection circuit 100. It should be noted that battery Bat or insulation detection circuit 100 may be a component of insulation resistance detection device 1.

[0018] Insulation resistance detection device 1 is mounted on, for example, a vehicle such as an electric vehicle that uses electric power for propulsion. High-voltage battery Bat is mounted on a vehicle such as an electric vehicle, and power is supplied from battery Bat to a drive load such as a motor to propel the vehicle. Battery Bat is a battery for an HV, PHEV, EV, or the like.

[0019] Insulation resistance refers to the insulation between a path through which a current flows and ground, and low insulation resistance can cause leakage current, which can lead to electric shock, fire, and the like. For this reason, detecting insulation resistance can detect dangerous vehicle conditions in advance. FIG. 1 shows insulation resistance Riso1 between the path connected to positive terminal t1 of battery Bat and ground GND, and insulation resistance Riso2 between the path connected to negative terminal t2 of battery Bat and ground GND. Ground GND is, for example, the potential of the vehicle chassis.

[0020] Insulation detection circuit 100 is a circuit for detecting insulation resistance Riso1 and Riso2, and includes a plurality of voltage dividing resistors connected between positive terminal t1 and negative terminal t2 of battery Bat. In Embodiment 1, resistors R1, R2, R3, R4, R5, and R6 are shown as the plurality of voltage dividing resistors. Specifically, resistors R1, R2, and R3 are connected in series between positive terminal t1 and ground GND, and resistors R4, R5, and R6 are connected in series between ground GND and negative terminal t2. In this way, positive terminal t1

is connected to ground GND via at least one resistor among the plurality of voltage dividing resistors, and negative terminal t2 is connected to ground GND via at least another resistor among the plurality of voltage dividing resistors. In Embodiment 1, the at least one resistor includes resistors R1, R2, and R3, and the at least another resistor includes resistors R4, R5, and R6. The potential of ground GND can be stabilized by connecting positive terminal t1 to ground GND and ground GND to negative terminal t2 via at least one resistor, respectively.

[0021] In addition, nodes N1, N2, N3, N4, and N5 are shown as the plurality of nodes between the plurality of voltage dividing resistors. Node N1 is a node between resistors R1 and R2, node N2 is a node between resistors R2 and R3, node N3 is a node between resistors R3 and R4, node N4 is a node between resistors R4 and R5, and node N5 is a node between resistors R5 and R6.

[0022] Although not shown here, insulation detection circuit 100 may include one or more switches connected to one or more of the plurality of voltage divider resistors. By controlling one or more switches included in insulation detection circuit 100, it is possible to change the voltage at any one of the plurality of nodes. Then, insulation resistance detection device 1 can accurately calculate the values of insulation resistance Riso1 and Riso2 from the voltages before and after the change.

[0023] It should be noted that insulation detection circuit 100 does not have to include one or more switches as shown in FIG. 1, and even in this case, it is possible to determine whether insulation resistance Riso1 or Riso2 is anomalous (i.e., whether a leakage current has occurred) from the voltage at any one of the plurality of nodes. This is because, if insulation resistance Riso1 or Riso2 is anomalous and its value is low, the voltage at any one of the plurality of nodes will also have an anomalous value.

[0024] Insulation resistance detection device 1 is a device for detecting insulation resistance Riso1 and Riso2, and includes voltage measurer 10 and controller 20. Insulation resistance detection device 1 is realized by, for example, a micro controller unit (MCU) or the like. In Embodiment 1, insulation resistance detection device 1 uses the potential of ground GND (node N3) as a reference potential.

[0025] Voltage measurer 10 is capable of measuring voltages at a plurality of nodes between a plurality of voltage dividing resistors included in insulation detection circuit 100. For example, voltage measurer 10 includes an A/D converter and measures the voltages at a plurality of nodes using the A/D converter. For example, voltage measurer 10 includes A/D converters 11 and 12 and is capable of measuring the voltage at node N1 using A/D converter 11 and measuring the voltage at node N2 using A/D converter 12. Node N1 is an example of a second node, and node N2 is an example of a first node. In Embodiment 1, voltage measurer 10 measures the voltages at nodes N1 and N2 using the potential of ground GND as a reference.

[0026] Controller 20 determines whether the voltage at node N2 measured by voltage measurer 10 is lower than or equal to a predetermined voltage. Controller 20 is an example of a determiner. The predetermined voltage is not particularly limited, and may be, for example, $\frac{1}{2}$, $\frac{1}{5}$, or the like of the absolute maximum rating of voltage measurer 10 (A/D converters 11 and 12).

[0027] In addition, when the voltage at node N2 measured by voltage measurer 10 is determined to be lower than or equal to a predetermined voltage, controller 20 switches a state of switch SW1 connected to node N1 among the plurality of nodes so that the voltage measured by voltage measurer 10 exceeds the predetermined voltage. Controller 20 is an example of a switcher.

[0028] In Embodiment 1, node N1 is a node to which a higher voltage is applied than node N2. Specifically, a voltage corresponding to two resistors R2 and R3 is applied to node N1, and a voltage corresponding to one resistor R3 is applied to node N2, so that a voltage higher than that at node N2 is applied to node N1.

[0029] In addition, in Embodiment 1, switch SW1 is connected between node N1 and voltage measurer 10, and when the voltage at node N2 measured by voltage measurer 10 is determined to be lower than or equal to a predetermined voltage, controller 20 switches a state of switch SW1 to connect node N1 and voltage measurer 10. Specifically, in this case, controller 20 switches the state

of switch SW1 from a non-conductive state to a conductive state. When the state of switch SW1 is switched to the conductive state, A/D converter 11 of voltage measurer 10 can measure the voltage at node N1.

[0030] In addition, controller 20 detects insulation resistance Riso1 and Riso2 based on the voltage measured by voltage measurer 10. For example, when the voltage measured by voltage measurer 10 is an anomalous value, controller 20 can determine that insulation resistance Riso1 or Riso2 is anomalous. For example, when controller 20 controls switch SW1 to a non-conductive state, it detects insulation resistance Riso1 and Riso2 using the voltage at node N2 measured by A/D converter 12, and when controller 20 controls switch SW1 to a conductive state, it detects insulation resistance Riso1 and Riso2 using the voltage at node N1 measured by A/D converter 11.

[0031] When insulation detection circuit 100 includes one or more switches, controller 20 can calculate the values of insulation resistance Riso1 and Riso2 using a predetermined algorithm from the voltages at node N1 or N2 measured before and after the states of one or more switches are switched.

[0032] It should be noted that voltage measurer 10 may include only one A/D converter, and the A/D converter may measure the voltages at nodes N1 and N2. In this case, a switch is provided between the A/D converter and node N2, and the voltage at node N2 is measured with the switch in a conductive state and switch SW1 in a non-conductive state. When the voltage at node N2 is determined to be lower than or equal to a predetermined voltage, the switch is made in a non-conductive state and switch SW1 is made in a conductive state, and the voltage at node N1 is measured.

[0033] In addition, controller 20 may diagnose a fault in insulation detection circuit 100 or voltage measurer 10 by comparing the voltage at node N2 with the voltage at node N1 which are measured by voltage measurer 10. Controller 20 is an example of a fault diagnoser. For example, the ratio between the voltage at node N1 and the voltage at node N2 should be a constant ratio according to the resistance values of resistors R1, R2, and R3, but if the constant ratio is not achieved, it can be determined that at least one of resistor R1, R2, or R3 is faulty, or that voltage measurer 10 is faulty.

[0034] In addition, four or more voltage dividing resistors may be connected between positive terminal t1 and ground GND, and for example, when the voltage at node N1 also falls below a predetermined voltage, a node to which a voltage higher than node N1 is applied may be connected by a switch to voltage measurer 10, and the voltage at that node may be measured. Then, when the voltage at that node also falls below the predetermined voltage, the voltage at another node to which a higher voltage is applied may be measured.

[0035] In addition, although not shown, voltage measurer 10 may measure the voltage at node N4 or N5, and controller 20 may detect insulation resistance Riso1 and Riso2 using the voltage at node N4 or N5 as well. Alternatively, the plurality of voltage dividing resistors may not be connected between ground GND and negative terminal t2 as shown in FIG. 1, and for example, only one resistor may be connected therebetween. In this case, voltage measurer 10 may not measure the voltage of the node between ground GND and negative terminal t2.

[0036] As described above, insulation resistance detection device 1 includes: voltage measurer 10 that measures a voltage at node N2 among a plurality of nodes between a plurality of voltage dividing resistors connected between positive terminal t1 and negative terminal t2 of battery Bat, the plurality of voltage dividing resistors being included in insulation detection circuit 100 for detecting insulation resistance Riso1 and Riso 2 in a path through which a current from battery Bat flows; a determiner (controller 20) that determines whether the voltage at node N2 measured by voltage measurer 10 is lower than or equal to a predetermined voltage; and a switcher (controller 20) that switches a state of switch SW1 connected to node N1 among the plurality of nodes to cause the voltage measured by voltage measurer 10 to exceed the predetermined voltage, when the voltage at node N2 measured by voltage measurer 10 is determined to be lower than or equal to the predetermined voltage.

[0037] According to this, when the voltage measured at node N2 falls below a predetermined voltage due to a voltage drop in battery Bat or during a measurement control process, the state of switch SW1 is switched so that the voltage measured by voltage measurer 10 exceeds the predetermined voltage. Therefore, voltage measurer 10 can measure a voltage higher than the predetermined voltage, and the measurement error of voltage measurer 10 can be reduced. Therefore, even if the voltage at node N2, where the voltage is measured, drops, a decrease in the detection accuracy of insulation resistance Riso1 and Riso2 can be suppressed. This will be explained in detail with reference to FIG. 2.

[0038] FIG. 2 is a diagram for explaining the effects of insulation resistance detection device 1 according to Embodiment 1.

[0039] As shown in (a) in FIG. 2, when the voltage measured by voltage measurer 10 (the voltage at node N2 in Embodiment 1) exceeds a predetermined voltage, switch SW1 is made in a non-conductive state. Thereafter, the voltage of battery Bat drops, and as shown in (b) in FIG. 2, it is assumed that the voltage measured by voltage measurer 10 (the voltage at node N2 in Embodiment 1) falls below the predetermined voltage. In this case, the ratio of error to the input voltage to voltage measurer 10 (A/D converter 11) increases, and the measurement error of voltage measurer 10 increases. Therefore, the state of switch SW1 connected between node N1 and voltage measurer 10 is switched from a non-conductive state to a conductive state, and voltage measurer 10 measures the voltage at node N1. As shown in (c) in FIG. 2, since the voltage at node N1 is higher than the voltage at node N2, the ratio of error to the input voltage to voltage measurer 10 (A/D converter 12) can be reduced, and the measurement error of voltage measurer 10 can be reduced.

[0040] It should be noted that voltage measurer 10 does not measure the voltage at node N1 before the voltage of battery Bat drops, in other words, before the voltage at node N2 falls below a predetermined voltage, because there is a risk that the voltage at node N1 will exceed the absolute maximum rating of voltage measurer 10.

[0041] For example, node N1 is a node to which a higher voltage is applied than node N2, switch SW1 is connected between node N1 and voltage measurer 10, and when the voltage at node N2 measured by voltage measurer 10 is determined to be lower than or equal to a predetermined voltage, the switcher (controller 20) switches a state of switch SW1 to connect node N1 and voltage measurer 10, and voltage measurer 10 may measure the voltage at node N1.

[0042] According to this, when the voltage measured at node N2 falls below a predetermined voltage due to a voltage drop in battery Bat or during a measurement control process, node N1, to which a voltage higher than that of node N2 is applied, is connected to voltage measurer 10 by switch SW1. Therefore, voltage measurer 10 can measure a voltage higher than the predetermined voltage at node N1, and the measurement error of voltage measurer 10 can be reduced. In this way, by switching the node at which voltage measurer 10 measures the voltage to a node to which a higher voltage is applied when the voltage of battery Bat drops or the like, a decrease in the detection accuracy of insulation resistance Riso1 and Riso2 can be suppressed.

[0043] For example, voltage measurer 10 may measure the voltages at nodes N1 and N2 using the potential of ground GND as a reference.

[0044] For example, insulation resistance detection device 1 is realized by a micro controller unit or the like, and in Embodiment 1, when the reference potential of the micro controller unit is set to the potential of ground GND, the voltages at nodes N1 and N2 can also be measured with ground GND potential as a reference.

[0045] For example, insulation resistance detection device 1 may further include a fault diagnoser (controller 20) that diagnoses a fault in insulation detection circuit 100 or voltage measurer 10 by comparing the voltage at node N2 with the voltage at node N1 which are measured by voltage measurer 10.

[0046] This makes it possible to diagnose a fault in insulation detection circuit 100 or voltage measurer 10 depending on whether the ratio of the voltage at node N1 to the voltage at node N2 is

constant.

(Variation of Embodiment 1)

[0047] Next, insulation resistance detection device **1a** according to a variation of Embodiment 1 will be described with reference to FIG. 3. Hereinafter, differences from insulation resistance detection device **1** according to Embodiment 1 will be mainly described, and descriptions of the same points will be omitted.

[0048] FIG. 3 is a configuration diagram showing an example of insulation resistance detection device **1a** according to a variation of Embodiment 1. It should be noted that in addition to insulation resistance detection device **1a**, FIG. 3 also shows battery Bat, insulation resistance Riso1 and Riso2 in a path through which current flows from battery Bat, and insulation detection circuit **100a**. It should be noted that battery Bat or insulation detection circuit **100a** may be a component of insulation resistance detection device **1a**.

[0049] Insulation resistance detection device **1a** is mounted on, for example, a vehicle, such as an electric vehicle, that uses electricity for propulsion.

[0050] Insulation detection circuit **100a** is a circuit for detecting insulation resistance Riso1 and Riso2, and includes a plurality of voltage dividing resistors connected between positive terminal t1 and negative terminal t2 of battery Bat. In the variation of Embodiment 1, resistors R7, R8, R9, and R10 are shown as the plurality of voltage dividing resistors. Specifically, resistor R7 is connected between positive terminal t1 and ground GND, and resistors R8, R9, and R10 are connected in series between ground GND and negative terminal t2. In this manner, positive terminal t1 is connected to ground GND via at least one resistor among the plurality of voltage dividing resistors, and negative terminal t2 is connected to ground GND via at least another resistor among the plurality of voltage dividing resistors. In the variation of Embodiment 1, the at least one resistor is resistor R7, and the at least another resistor includes resistors R8, R9, and R10. The potential of ground GND can be stabilized by connecting positive terminal t1 to ground GND and ground GND to negative terminal t2 via at least one resistor, respectively.

[0051] In addition, nodes N6, N7, and N8 are shown as a plurality of nodes between the voltage dividing resistors. Node N6 is a node between resistors R7 and R8, node N7 is a node between resistors R8 and R9, and node N8 is a node between resistors R9 and R10.

[0052] Although not shown here, insulation detection circuit **100a** may include one or more switches connected to one or more of the plurality of voltage divider resistors. By controlling one or more switches included in insulation detection circuit **100a**, the voltage at any one of the plurality of nodes can be changed. Then, insulation resistance detection device **1a** can accurately calculate the values of insulation resistance Riso1 and Riso2 from the voltages before and after the change.

[0053] It should be noted that insulation detection circuit **100a** may not include one or more switches as shown in FIG. 3, and even in this case, it is possible to determine whether insulation resistance Riso1 or Riso2 is anomalous (i.e., whether a leakage current has occurred) from the voltage at any one of the plurality of nodes.

[0054] Insulation resistance detection device **1a** is a device for detecting insulation resistance Riso1 and Riso2, and includes voltage measurer **10a** and controller **20a**. Insulation resistance detection device **1a** is realized by, for example, a micro controller unit or the like. In the variation of Embodiment 1, insulation resistance detection device **1a** uses the potential of negative terminal t2 as a reference potential.

[0055] Voltage measurer **10a** can measure voltages at a plurality of nodes between a plurality of voltage dividing resistors included in insulation detection circuit **100a**. For example, voltage measurer **10a** includes an A/D converter and measures voltages at the plurality of nodes by the A/D converter. For example, voltage measurer **10a** includes A/D converters **11a** and **12a** and can measure the voltage at node N7 by A/D converter **11a** and can measure the voltage at node N8 by A/D converter **12a**. Node N7 is an example of a second node, and node N8 is an example of a first

node. In the variation of Embodiment 1, voltage measurer **10a** measures the voltages at nodes **N7** and **N8** using the potential of negative terminal **t2** as a reference.

[0056] Controller **20a** determines whether the voltage at node **N8** measured by voltage measurer **10a** is lower than or equal to a predetermined voltage. Controller **20a** is an example of a determiner. The predetermined voltage is not particularly limited, but may be, for example, $\frac{1}{2}$, $\frac{1}{5}$, or the like of the absolute maximum rating of voltage measurer **10a** (A/D converters **11a** and **12a**).

[0057] In addition, when controller **20a** determines that the voltage at node **N8** measured by voltage measurer **10a** is lower than or equal to a predetermined voltage, controller **20a** switches a state of switch **SW1a** connected to node **N7** among the plurality of nodes so that the voltage measured by voltage measurer **10a** exceeds the predetermined voltage. Controller **20a** is an example of a switcher. Node **N7** is an example of a second node.

[0058] In the variation of Embodiment 1, node **N7** is a node to which a higher voltage than that of node **N8** is applied. Specifically, a voltage corresponding to two resistors **R9** and **R10** is applied to node **N7**, and a voltage corresponding to single resistor **R10** is applied to node **N8**, so that a voltage higher than that of node **N8** is applied to node **N7**.

[0059] In addition, in the variation of Embodiment 1, switch **SW1a** is connected between node **N7** and voltage measurer **10a**, and when the voltage at node **N8** measured by voltage measurer **10a** is determined to be lower than or equal to a predetermined voltage, controller **20a** switches a state of switch **SW1a** to connect node **N7** and voltage measurer **10a**. Specifically, in this case, controller **20a** switches the state of switch **SW1a** from a non-conductive state to a conductive state. When the state of switch **SW1a** is switched to the conductive state, A/D converter **11a** of voltage measurer **10a** can measure the voltage at node **N7**.

[0060] In addition, controller **20a** detects insulation resistance **Riso1** and **Riso2** based on the voltage measured by voltage measurer **10a**. For example, when the voltage measured by voltage measurer **10a** is an anomalous value, controller **20a** can determine that insulation resistance **Riso1** or **Riso2** is anomalous. For example, when controller **20a** controls switch **SW1a** to a non-conductive state, it detects insulation resistance **Riso1** and **Riso2** using the voltage at node **N8** measured by A/D converter **12a**, and when controller **20a** controls switch **SW1a** to a conductive state, it detects insulation resistance **Riso1** and **Riso2** using the voltage at node **N7** measured by A/D converter **11a**.

[0061] When insulation detection circuit **100a** includes one or more switches, controller **20a** can calculate the values of insulation resistance **Riso1** and **Riso2** using a predetermined algorithm from the voltages at node **N7** or **N8** measured before and after the states of one or more switches are switched.

[0062] It should be noted that voltage measurer **10a** may include only one A/D converter, and the A/D converter may measure the voltages at nodes **N7** and **N8**. In this case, a switch is also provided between the A/D converter and node **N8**, and the voltage at node **N8** is measured with the switch in a conductive state and switch **SW1a** in a non-conductive state. When the voltage at node **N8** is determined to be lower than or equal to a predetermined voltage, the switch is made in a non-conductive state and switch **SW1a** is made in a conductive state, and the voltage at node **N7** is measured.

[0063] In addition, controller **20a** may diagnose a fault in insulation detection circuit **100a** or voltage measurer **10a** by comparing the voltage at node **N8** with the voltage at node **N7** which are measured by voltage measurer **10a**. Controller **20a** is an example of a fault diagnoser. For example, the ratio between the voltage at node **N7** and the voltage at node **N8** should be a constant ratio according to the resistance values of resistors **R8**, **R9**, and **R10**, but if the constant ratio is not achieved, it can be determined that at least one of resistor **R8**, **R9**, or **R10** is faulty, or that voltage measurer **10a** is faulty.

[0064] In addition, four or more voltage dividing resistors may be connected between ground **GND** and negative terminal **t2**, and for example, when the voltage at node **N7** also falls below a

predetermined voltage, a node to which a voltage higher than that of node N7 is applied may be connected by a switch to voltage measurer **10a**, and the voltage at that node may be measured. Then, when the voltage at that node also falls below the predetermined voltage, the voltage at another node to which a higher voltage is applied may be measured.

[0065] As described above, in the variation of Embodiment 1, voltage measurer **10a** may measure the voltages at nodes N7 and N8 using the potential of negative terminal t2 as a reference.

[0066] For example, insulation resistance detection device **1a** is realized by a micro controller unit or the like, and in the variation of Embodiment 1, when the reference potential of the micro controller unit is set to the potential of negative terminal t2, the voltages at nodes N7 and N8 can also be measured using the potential of negative terminal t2 as a reference.

[0067] It should be noted that in the variation of Embodiment 1, as in Embodiment 1, even if the voltage of node N8 drops due to a voltage drop in battery Bat or during a measurement control process, a decrease in the detection accuracy of insulation resistance Riso1 and Riso2 can be suppressed.

Embodiment 2

[0068] Next, insulation resistance detection device **1b** according to Embodiment 2 will be described with reference to FIG. 4. Hereinafter, differences from insulation resistance detection device **1** according to Embodiment 1 will be mainly described, and descriptions of the same points will be omitted.

[0069] FIG. 4 is a configuration diagram showing an example of insulation resistance detection device **1b** according to Embodiment 2. It should be noted that in addition to insulation resistance detection device **1b**, FIG. 4 also shows battery Bat, insulation resistance Riso1 and Riso2 in a path through which current flows from battery Bat, and insulation detection circuit **100b**. It should be noted that battery Bat or insulation detection circuit **100b** may be a component of insulation resistance detection device **1b**.

[0070] Insulation resistance detection device **1b** is mounted on a vehicle, such as an electric vehicle, that uses electricity for propulsion.

[0071] Insulation detection circuit **100b** is a circuit for detecting insulation resistance Riso1 and Riso2, and includes a plurality of voltage dividing resistors connected between positive terminal t1 and negative terminal t2 of battery Bat. In Embodiment 2, resistors R11, R12, R13, R14, and R15 are shown as the plurality of voltage dividing resistors. Specifically, resistors R11 and R14 are connected in parallel between positive terminal t1 and ground GND, and a circuit in which resistors R12 and R15 are connected in parallel and resistor R13 are connected in series between ground GND and negative terminal t2. In this way, positive terminal t1 is connected to ground GND via at least one resistor among the plurality of voltage dividing resistors, and negative terminal t2 is connected to ground GND via at least another resistor among the plurality of voltage dividing resistors. In Embodiment 2, the at least one resistor includes resistors R11 and R14, and the at least another resistor includes resistors R12, R13, and R15. The potential of ground GND can be stabilized by connecting positive terminal t1 to ground GND, and ground GND to negative terminal t2 via one or more resistors, respectively.

[0072] In addition, nodes N9 and N10 are shown as a plurality of nodes between the voltage dividing resistors. Node N9 is a node between the parallel circuit of resistors R11 and R14 and the parallel circuit of resistors R12 and R15, and node N10 is a node between the parallel circuit of resistors R12 and R15 and resistor R13.

[0073] Insulation detection circuit **100b** includes switches SW2 and SW3. Switches SW2 and SW3 are connected to node N9 and one of the plurality of voltage dividing resistors. Here, switch SW2 is connected between node N9 and resistor R14, and switch SW3 is connected between node N9 and resistor R15. Node N9 is an example of a second node. When switch SW2 is in a conductive state, the resistance value of the parallel circuit of resistors R11 and R14 is small, and when switch SW2 is in a non-conductive state, the resistance value of the parallel circuit of resistors R11 and

R14 is large. When switch **SW3** is in a conductive state, the resistance value of the parallel circuit of resistors **R12** and **R15** is small, and when switch **SW3** is in a non-conductive state, the resistance value of the parallel circuit of resistors **R12** and **R15** is large.

[0074] Although not shown here, in addition to switches **SW2** and **SW3**, insulation detection circuit **100b** may further include one or more switches connected to one or more voltage dividing resistors among resistors **R11**, **R12**, and **R13**. The voltage at node **N10** can be changed by controlling the one or more switches further included in insulation detection circuit **100b**. Then, the values of insulation resistance **Riso1** and **Riso2** can be accurately calculated by insulation resistance detection device **1b** from the voltages before and after the change.

[0075] It should be noted that as shown in FIG. 4, insulation detection circuit **100b** may not include any switches other than switches **SW2** and **SW3**, and even in this case, it is possible to determine from the voltage at node **N10** whether insulation resistance **Riso1** or **Riso2** is anomalous (i.e., whether a leakage current has occurred).

[0076] Insulation resistance detection device **1b** is a device for detecting insulation resistance **Riso1** and **Riso2**, and includes voltage measurer **10b** and controller **20b**. Insulation resistance detection device **1b** is realized by, for example, a micro controller unit. In Embodiment 2, insulation resistance detection device **1b** uses the potential of negative terminal **t2** as a reference potential.

[0077] Voltage measurer **10b** measures the voltage at node **N10** among a plurality of nodes between a plurality of voltage dividing resistors included in insulation detection circuit **100b**. For example, voltage measurer **10b** includes A/D converter **13**, and measures the voltage at node **N10** using A/D converter **13**. Node **N10** is an example of a first node. In Embodiment 2, voltage measurer **10b** measures the voltage at node **N10** using the potential of negative terminal **t2** as a reference.

[0078] Controller **20b** determines whether the voltage at node **N10** measured by voltage measurer **10b** is lower than or equal to a predetermined voltage. Controller **20b** is an example of a determiner. The predetermined voltage is not particularly limited, but may be, for example, $\frac{1}{2}$, $\frac{1}{5}$, or the like of the absolute maximum rating of voltage measurer **10b** (A/D converter **13**).

[0079] In addition, when the voltage at node **N10** measured by voltage measurer **10b** is determined to be lower than or equal to a predetermined voltage, controller **20b** switches states of switches **SW2** and **SW3** connected to node **N9** so that the voltage at node **N10** measured by voltage measurer **10b** exceeds the predetermined voltage. Controller **20b** is an example of a switcher.

[0080] For example, when switches **SW2** and **SW3** are in a non-conductive state and the voltage at node **N10** is lower than or equal to a predetermined voltage, controller **20b** switches a state of at least one of switch **SW2** or **SW3** to a conductive state. This allows the voltage at node **N10** to be increased, and the voltage at node **N10** can exceed the predetermined voltage.

[0081] In Embodiment 1, an example was described in which the node at which voltage measurer **10** measures the voltage is switched by controlling switch **SW1**, but in Embodiment 2, the voltage applied to the node at which voltage measurer **10b** measures the voltage is switched by controlling switches **SW2** and **SW3**.

[0082] In addition, controller **20b** detects insulation resistance **Riso1** and **Riso2** based on the voltage measured by voltage measurer **10b**. For example, when the voltage measured by voltage measurer **10b** is an anomalous value, controller **20b** can determine that insulation resistance **Riso1** or **Riso2** is anomalous.

[0083] It should be noted that when insulation detection circuit **100b** includes one or more switches in addition to switches **SW2** and **SW3**, controller **20b** can calculate the values of insulation resistance **Riso1** and **Riso2** using a predetermined algorithm from the voltages at node **N10** measured before and after the states of one or more switches are switched.

[0084] In addition, controller **20b** may diagnose a fault in insulation detection circuit **100b** or voltage measurer **10b** by comparing the voltages at node **N10** measured by voltage measurer **10b** before and after the states of switches **SW2** and **SW3** are switched. Controller **20b** is an example of a fault diagnoser. For example, the ratio between the voltages at node **N10** before and after the

states of switches SW2 and SW3 are switched should be a constant ratio according to the resistance values of resistors R11, R12, R13, R14, and R15, but if the constant ratio is not achieved, it can be determined that at least one of resistor R11, R12, R13, R14, or R15 is faulty, or that voltage measurer 10b is faulty.

[0085] In addition, the switch connected to node N9 includes two switches SW2 and SW3 has been described in Embodiment 2, but the present invention is not limited thereto. For example, the switch connected to node N9 may be one, or three or more.

[0086] In addition, an example in which voltage measurer 10b measures the voltage at node N10 using the potential of negative terminal t2 as a reference has been described in Embodiment 2, but voltage measurer 10b may also measure the voltage at node N10 using the potential of ground GND as a reference.

[0087] The effects of insulation resistance detection device 1b according to Embodiment 2 can also be explained with reference to FIG. 2, in the same manner as Embodiment 1.

[0088] As shown in (a) in FIG. 2, when the voltage measured by voltage measurer 10b (the voltage at node N10 in Embodiment 2) exceeds a predetermined voltage, switches SW2 and SW3 are made in a non-conductive state. Thereafter, the voltage of battery Bat drops, and as shown in (b) in FIG. 2, it is assumed that the voltage measured by voltage measurer 10b (the voltage at node N10 in Embodiment 2) falls below the predetermined voltage. In this case, the ratio of error to the input voltage to voltage measurer 10b (A/D converter 13) increases, and the measurement error of voltage measurer 10b increases. Therefore, the states of switches SW2 and SW3 are switched from a non-conductive state to a conductive state, and voltage measurer 10b measures the voltage at node N10. As shown in (c) in FIG. 2, since the voltage at node N10 is higher than the voltage before the states of switches SW2 and SW3 are switched, the ratio of error to the input voltage to voltage measurer 10b (A/D converter 13) can be reduced, and the measurement error of voltage measurer 10b can be reduced.

[0089] It should be noted that also in Embodiment 2, as in Embodiment 1, a switch may be connected between a node to which a voltage higher than node N10 is applied and voltage measurer 10b, and when the voltage at node N10 is determined to be lower than or equal to a predetermined voltage, the state of the switch may be switched to measure the voltage at the node to which a voltage higher than node N10 is applied. This can improve the detection accuracy of insulation resistance Riso1 and Riso2.

[0090] As described above, in Embodiment 2, switches SW2 and SW3 are connected to node N9 and any one of a plurality of voltage dividing resistors, and when the voltage at node N10 measured by voltage measurer 10b is determined to be lower than or equal to a predetermined voltage, the switcher (controller 20b) switches the states of switches SW2 and SW3 so that the voltage at node N10 measured by voltage measurer 10b exceeds the predetermined voltage.

[0091] According to this, when the voltage measured at node N10 falls below a predetermined voltage due to a voltage drop in battery Bat or during a measurement control process, the states of switches SW2 and SW3 are switched so that the voltage applied to node N10 becomes higher. Therefore, voltage measurer 10b can measure a voltage higher than the predetermined voltage at node N10, and the measurement error of voltage measurer 10b can be reduced. In this way, by switching the voltage applied to node N10 at which voltage measurer 10b measures the voltage when the voltage of battery Bat drops or the like, a decrease in the detection accuracy of insulation resistance Riso1 and Riso2 can be suppressed.

[0092] For example, voltage measurer 10b may measure the voltage at node N10 using the potential of ground GND as a reference.

[0093] For example, insulation resistance detection device 1b is realized by a micro controller unit or the like, and in Embodiment 2, when the reference potential of the micro controller unit is set to ground GND potential, the voltage at node N10 can also be measured using the potential of ground GND as a reference.

[0094] Alternatively, for example, voltage measurer **10b** may measure the voltage at node **N10** using the potential of negative terminal **t2** as a reference.

[0095] For example, insulation resistance detection device **1b** is realized by a micro controller unit or the like, and in Embodiment 2, when the reference potential of the micro controller unit is set to the potential of negative terminal **t2**, the voltage at node **N10** can also be measured using the potential of negative terminal **t2** as a reference.

[0096] For example, insulation resistance detection device **1b** may further include a fault diagnoser (controller **20b**) that diagnoses a fault in insulation detection circuit **100b** or voltage measurer **10b** by comparing the voltages at node **N10** measured by voltage measurer **10b** before and after the states of switches **SW2** and **SW3** are switched.

[0097] This makes it possible to diagnose a fault in insulation detection circuit **100b** or voltage measurer **10b** depending on whether the ratio of the voltages at node **N10** before and after the states of switches **SW2** and **SW3** are switched is constant.

OTHER EMBODIMENTS

[0098] As described above, the embodiments have been described as examples of the technology according to the present disclosure. However, the technology according to the present disclosure is not limited thereto, and can be applied to embodiments in which modifications, substitutions, additions, omissions, or the like are made as appropriate. For example, the following variations are also included in one embodiment of the present disclosure.

[0099] For example, the present disclosure can be realized not only as an insulation resistance detection device, but also as an insulation resistance detection method that includes steps (processing) performed by components included in the insulation resistance detection device.

[0100] FIG. 5 is a flowchart showing an example of an insulation resistance detection method according to another embodiment.

[0101] As shown in FIG. 5, the insulation resistance detection method includes: a voltage measuring step (step **S11**) of measuring a voltage at a first node among a plurality of nodes between a plurality of voltage dividing resistors connected between a positive terminal and a negative terminal of a battery, the plurality of voltage dividing resistors being included in an insulation detection circuit for detecting insulation resistance in a path through which a current from the battery flows; a determining step (step **S12**) of determining whether the voltage at the first node measured in the measuring is lower than or equal to a predetermined voltage; and a switching step (step **S13**) of switching a state of a switch connected to a second node among the plurality of nodes to cause the voltage measured in the measuring to exceed the predetermined voltage, when the voltage at the first node measured in the measuring is determined to be lower than or equal to the predetermined voltage (Yes in step **S12**).

[0102] For example, the steps in the insulation resistance detection method may be executed by a computer (computer system). The present disclosure can be realized as a program for causing a computer to execute the steps included in the insulation resistance detection method.

[0103] Furthermore, the present disclosure can be realized as a non-transitory computer-readable recording medium such as a CD-ROM having recorded the program thereon.

[0104] For example, when the present invention is realized as a program (software), each step is executed by running the program using hardware resources such as a computer's CPU, memory, input/output circuits, and the like. That is, each step is executed by the CPU obtaining data from memory, input/output circuits, or the like to perform calculations, or outputting the results of the calculations to memory, input/output circuits, or the like.

[0105] In addition, each component included in the insulation resistance detection devices of the above embodiments may be realized as a dedicated or general-purpose circuit.

[0106] In addition, each component included in the insulation resistance detection devices of the above embodiments may be realized as a large scale integration (LSI), which is an integrated circuit (IC).

[0107] In addition, the integrated circuit is not limited to an LSI, and may be realized by a dedicated circuit or a general-purpose processor. A programmable field programmable gate array (FPGA) or a reconfigurable processor in which the connections and settings of circuit cells within the LSI can be reconfigured may be used.

[0108] Furthermore, if an integrated circuit technology that replaces an LSI appears due to advances in semiconductor technology or another technology derived therefrom, that technology may naturally be used to integrate the components included in the insulation resistance detection device.

[0109] In addition, forms obtained by applying various modifications to each embodiment conceived by a person skilled in the art or forms realized by arbitrarily combining the components and functions in each embodiment without departing from the spirit of the present disclosure are also included in this disclosure.

INDUSTRIAL APPLICABILITY

[0110] The present disclosure can be applied to a device that detects insulation resistance in a path through which current flows from a high-voltage battery mounted in a vehicle or the like.

REFERENCE SIGNS LIST

[0111] **1, 1a, 1b** Insulation resistance detection device [0112] **10, 10a, 10b** Voltage measurer [0113] **11, 11a, 12, 12a, 13** A/D converter [0114] **20, 20a, 20b** Controller [0115] **100, 100a, 100b** Insulation detection circuit [0116] **Bat** Battery [0117] **GND** Ground [0118] **N1, N2, N3, N4, N5, N6, N7, N8, N9, N10** Node [0119] **R1, R2, R3, R4, R5, R6, R7, R8, R9, R10, R11, R12, R13, R14, R15** Resistance [0120] **Riso1, Riso2** Insulation resistance [0121] **SW1, SW1a, SW2, SW3** Switch [0122] **t1** Positive terminal [0123] **t2** Negative terminal

Claims

1. An insulation resistance detection device comprising: a voltage measurer that measures a voltage at a first node among a plurality of nodes between a plurality of voltage dividing resistors connected between a positive terminal and a negative terminal of a battery, the plurality of voltage dividing resistors being included in an insulation detection circuit for detecting insulation resistance in a path through which a current from the battery flows; a determiner that determines whether the voltage at the first node measured by the voltage measurer is lower than or equal to a predetermined voltage; and a switcher that switches a state of a switch connected to a second node among the plurality of nodes to cause the voltage measured by the voltage measurer to exceed the predetermined voltage, when the voltage at the first node measured by the voltage measurer is determined to be lower than or equal to the predetermined voltage.
2. The insulation resistance detection device according to claim 1, wherein the second node is a node to which a voltage higher than the voltage at the first node is applied, the switch is connected between the second node and the voltage measurer, the switcher switches the state of the switch to connect the second node and the voltage measurer, when the voltage at the first node measured by the voltage measurer is determined to be lower than or equal to the predetermined voltage, and the voltage measurer measures a voltage at the second node.
3. The insulation resistance detection device according to claim 2, wherein the positive terminal is connected to ground via at least one of the plurality of voltage dividing resistors, the negative terminal is connected to the ground via at least an other one of the plurality of voltage dividing resistors, and the voltage measurer measures the voltage at the first node and the voltage at the second node using a potential of the ground as a reference.
4. The insulation resistance detection device according to claim 2, wherein the voltage measurer measures the voltage at the first node and the voltage at the second node using a potential of the negative terminal as a reference.
5. The insulation resistance detection device according to claim 2, further comprising: a fault

- diagnoser that diagnoses a fault in the insulation detection circuit or the voltage measurer by comparing the voltage at the first node with the voltage at the second node which are measured by the voltage measurer.
- 6.** The insulation resistance detection device according to claim 1, wherein the switch is connected to the second node and any one of the plurality of voltage dividing resistors, and the switcher switches the state of the switch to cause the voltage at the first node measured by the voltage measurer to exceed the predetermined voltage, when the voltage at the first node measured by the voltage measurer is determined to be lower than or equal to the predetermined voltage.
- 7.** The insulation resistance detection device according to claim 6, wherein the positive terminal is connected to ground via at least one of the plurality of voltage dividing resistors, the negative terminal is connected to the ground via at least an other one of the plurality of voltage dividing resistors, and the voltage measurer measures the voltage at the first node using a potential of the ground as a reference.
- 8.** The insulation resistance detection device according to claim 6, wherein the voltage measurer measures the voltage at the first node using a potential of the negative terminal as a reference.
- 9.** The insulation resistance detection device according to claim 6, further comprising: a fault diagnoser that diagnoses a fault in the insulation detection circuit or the voltage measurer by comparing the voltage at the first node measured by the voltage measurer before the state of the switch is switched and the voltage at the first node measured by the voltage measurer after the state of the switch is switched.
- 10.** An insulation resistance detection method comprising: measuring a voltage at a first node among a plurality of nodes between a plurality of voltage dividing resistors connected between a positive terminal and a negative terminal of a battery, the plurality of voltage dividing resistors being included in an insulation detection circuit for detecting insulation resistance in a path through which a current from the battery flows; determining whether the voltage at the first node measured in the measuring is lower than or equal to a predetermined voltage; and switching a state of a switch connected to a second node among the plurality of nodes to cause the voltage measured in the measuring to exceed the predetermined voltage, when the voltage at the first node measured in the measuring is determined to be lower than or equal to the predetermined voltage.
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